$$= uv_{(1)} - u^{(1)}v_{(2)} + \int u^{(2)}dv_{(3)}$$

$$= uv_{(1)} - u^{(1)}v_{(2)} + \left(u^{(2)}v_{(3)} - \int v_{(3)}du^{(2)}\right)$$

$$= uv_{(1)} - u^{(1)}v_{(2)} + u^{(2)}v_{(3)} - \int v_{(3)}du^{(2)} \ .$$

Proceeding in this way, we get

$$\int uvdx = uv_{(1)} - u^{(1)}v_{(2)} + u^{(2)}v_{(3)} - u^{(3)}v_{(4)} + \cdots .$$

The above result is called the **Bernoulli's formula for integration of product of two functions.**

Note

Since u is a polynomial function of x, the successive derivative $u^{(m)}$ will be zero for some positive integer m and so all further derivatives will be zero only. Hence the right-hand-side of the above formula contains a finite number of terms only.

Example 9.31

Evaluate $\int_0^{\pi} x^2 \cos nx\, dx$, where n is a positive integer.

Solution

Taking $u = x^2$ and $v = \cos nx$, and applying the Bernoulli's formula, we get

$$I = \int_0^{\pi} x^2 \cos nx\, dx = \left[\left(x^2\right)\left(\frac{\sin nx}{n}\right) - (2x)\left(-\frac{\cos nx}{n^2}\right) + (2)\left(-\frac{\sin nx}{n^3}\right)\right]_0^{\pi}$$

$$= \frac{2\pi(-1)^n}{n^2}, \text{ since } \cos n\pi = (-1)^n \text{ and } \sin n\pi = 0 .$$

Example 9.32

Evaluate : $\int_0^1 e^{-2x}(1 + x - 2x^3)\, dx$.

Solution

Taking $u = 1 + x - 2x^3$ and $v = e^{-2x}$, and applying the Bernoulli's formula, we get

$$I = \int_0^1 e^{-2x}(1 + x - 2x^3)\, dx$$

$$= \left[(1 + x - 2x^3)\left(\frac{e^{-2x}}{-2}\right) - \left(1 - 6x^2\right)\left(\frac{e^{-2x}}{4}\right) + (-12x)\left(\frac{e^{-2x}}{-8}\right) - (-12)\left(\frac{e^{-2x}}{16}\right)\right]_0^1$$

$$= \left[\frac{e^{-2x}}{16}\left(16x^3 + 24x^2 + 16x\right)\right]_0^1$$

$$= \frac{7}{2e^2} .$$

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